Tutorial-1

1) A Smart Farming system protects whether a plant will grow well(1) or poorly(0) based on daily sunlight hours and water amount in liters.

Dataset:

Sunlight Hours (x_1)	Water Liters (x_2)	Bias (x_3)	Target (Good growth=1, Poor=0)
3	1	1	0
8	4	1	1
5	2	1	0
10	5	1	1
7	3	1	1

1) Mean of x_1

$$\mu x_1 = \frac{3+8+5+10+7}{5}$$

$$= \frac{33}{5} = 6.6$$

2) Mean of x_2

$$\mu x_2 = \frac{1+4+2+5+3}{5}$$

$$= \frac{15}{5} = 3$$

$$* \text{Variance} = \sigma^2 = \frac{1}{N} \sum_i (x_i - \mu)^2$$

N → Total data Inputs

x_i → Input values

μ → Mean/Average

$$Z = W_1X_1 + W_2X_2 + W_3X_3 + b$$

$$Z \Rightarrow \text{if } Z \geq 0, 1$$

$$Z \leq 0, 0$$

$$\boxed{Z = W_1X_1 + W_2X_2 + W_3X_3}$$

Initial weights $W_1 = W_2 = W_3 = 0$ $\eta = 0.1$

row1 (3, 1, 1)

$$Z_1 = (0)(3) + (0)(1) + (0)(1)$$
$$= 0$$
$$Z_1 = \hat{y}_1 = 0$$
$$\hat{y}_1 = 1$$

Here $\hat{y}_1 = 1$ and $y = 0$ $\boxed{\hat{y}_1 \neq \hat{y}}$

So, update weights

$$w_1^{new} = w_1^{old} + \eta (y - \hat{y}) x_i$$

$$w_1^{new} = 0 + 0.1(0-1)3$$
$$= 0.1(-3) = -0.3$$
$$w_2^{new} = 0 + 0.1(0-1)(1)$$
$$= -0.1$$
$$w_3^{new} = 0 + 0.1(0-1)(1)$$
$$= -0.1$$

row2 (8, 4, 1)

$w_1 = 0.3$

$w_2 = -0.1$ $\quad \eta = 0.1$

$w_3 = -0.1$

$$Z_2 = X_1 W_1 + X_2 W_2 + X_3 W_3$$
$$= (-0.3)(8) + (-0.1)(4) + (-0.1)(1)$$
$$= -2.9$$

$\hat{y} = Z_2 < 0$ $\quad \boxed{\hat{y} = 0}$ $\quad \boxed{y = 1}$

Here $\hat{y} \neq y$, update weights

$$w_1^{new} = w_1^{old} + \alpha (y - \hat{y}) x_1$$
$$w_1^{new} = -0.3 + 0.1(1-0)(8)$$
$$= 0.5$$

$$w_2^{new} = -0.1 + 0.1(1-0)(1)$$
$$= 0$$
$$w_3^{new} = -0.1 + 0.1(1-0)(1)$$
$$= 0$$

row 3 (5, 2, 1)

$w_1 = 0.5$

$w_2 = 0.3$ $\quad \eta = 0.1$

$w_3 = 0$

$$z_3 = w_1x_1 + w_2x_2 + w_3x_3$$
$$= 0.5(5) + 0.3(2) + 0(1)$$
$$= 3.1$$

$\hat{y} = z_3 > 0, \ \hat{y} = 1, \ y = 0$

$\therefore$ Here $\hat{y} \neq y$, update weights

$$w_i^{new} = w_i^{old} + \eta(y - \hat{y}) \times 1$$

$$w_1^{new} = 0.5 + 0.1(0-1)5$$
$$= 0$$
$$w_2^{new} = 0.3 + 0.1(0-1)2$$
$$= 0.1$$
$$w_3^{new} = 0 + 0.1(0-1)1$$
$$= -0.1$$

row$_4$ (10, 5, 1)

$w_1 = 0$

$w_2 = 0.1$ $\quad \eta = 0.1 \quad y = 1$

$w_3 = -0.1$

$$z_4 = w_1x_1 + w_2x_2 + w_3x_3$$
$$= 10(0) + 0.1(5) + 0.1(1)$$
$$= 0.4$$

$\hat{y} = z_4 > 0$ $\quad \boxed{\hat{y} = 1}$ $\quad \boxed{y = \hat{y}}$ No weight updation

Row 5 (7, 3, 1)

$w_1 = 0$

$w_2 = 0.1$

$w_3 = -0.1$

$\eta = 0.1$

$y = 1$

$z_5 = w_1x_1 + w_2x_2 + w_3x_3$

$= 0(7) + 0.1(3) + (-0.1)(1)$

$= 0.2$

$\hat{y} = z_5 \geq 0$

$\boxed{\hat{y} = 1}$

$\therefore$ No Weight Updation

$\therefore \boxed{y = \hat{y}}$